

Shashwat Revdandekar

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Education

Georgia Institute of Technology

Master of Science in Electrical and Computer Engineering

Atlanta, GA Aug 2025 - Dec 2026

GPA: 3.75/4.0

The Maharaja Sayajirao University of Baroda

Bachelor of Engineering in Electronics Engineering

Vadodara, India Jul 2020 - May 2024

GPA: 4.0/4.0

Skills

- **Computer Architecture:** Microarchitecture, Pipeline Design, Out-of-Order Execution, Memory Hierarchy, Cache Design
- **Programming Languages:** C/C++, Python, MATLAB, Verilog/VHDL, Assembly
- **Hardware & EDA:** KICAD, Proteus, PCB Design, Cadence Virtuoso, HSPICE, Xilinx Vivado, HFSS
- **Testing and Lab Equipment:** Digital Oscilloscopes, Multimeters, Network Analyzers, Spectrum Analyzers, Logic Analyzers

Research Projects

IGZO-Based 2T0C DRAM as an Intrinsic Rowhammer Mitigation: A Device Physics Analysis

[PDF]

- Analyzed IGZO-based 2T0C DRAM as an intrinsic Rowhammer mitigation, showing its ultra-low off-state leakage and elimination of the storage capacitor jointly remove the charge-disturbance path exploited by classical Rowhammer attacks
- Proposed 2T0C IGZO DRAM as a materials-driven Rowhammer defense that requires no software or firmware mitigation layer, contrasting with system-level approaches like Target Row Refresh (TRR)
- Benchmarked findings against published Rowhammer thresholds across DDR3-DDR5, evaluating adoption tradeoffs

Academic & Hardware Projects

Microarchitecture Design: Superscalar Pipeline, GEMM Optimization and Systolic Arrays

- Built a systolic-array GEMM simulator to study cycles-to-completion, FIFO sizing, and off-chip bandwidth scaling
- Developed a cycle-accurate C++ simulator modeling superscalar out-of-order execution with RAT and ROB
- Implemented pipeline forwarding, branch prediction, and hazard resolution to model high-performance CPU execution
- Evaluated architectural performance using CPI, pipeline utilization, and branch misprediction metrics across instruction traces

Efficient Delay Fault Simulation for Short Defects in Standard Cells

- Conducted transistor-level reliability analysis of resistive shorts in CMOS standard cells to model defect-induced timing failures
- Characterized standard cell delay under varying defect resistances to support timing margin requirements
- Designed Cadence testbenches and executed HSPICE simulations to analyze defect behavior at transistor-level fault locations
- Developed Python scripts to dynamically modify netlist resistance values, execute HSPICE simulations, and extract results

Realization of Digital Circuits on Artix-7 FPGA Board

- Developed VHDL modules using Xilinx Vivado for the full design flow from RTL synthesis to bitstream generation
- Authored functional verification testbenches, ensuring cycle-accurate behavior and complete logic coverage before deployment
- Performed timing analysis and resource checks to optimize design footprint and meet clock frequency constraints

AI-based Industrial Robotic Arm

- Developed an object classification model integrated with a custom robotic arm prototype for automated manipulation tasks.
- Configured Raspberry Pi-based camera and sensor integration to provide feedback control and environment sensing capabilities.
- Developed an embedded C firmware for the ARM-based M051 to control the robotic arm movement via I2C protocol

Professional Experience

Energy Materials Lab, Georgia Tech | Graduate Assistant - PCB Design

Atlanta, GA | Sep 2025 – Present

- Engineered a multi-channel PCB in KiCAD enabling 8-pixel measurement, reducing characterization time by almost 80%
- Designed interfaces for direct connectivity with legacy hardware, ensuring complete software and hardware compatibility
- Managed the entire PCB fabrication and assembly process with external vendors, including DFM and DRC reviews
- Optimized layouts to prevent shading while maintaining Signal Integrity (SI) for high-precision, low-noise data collection

Adani Solar | Process Engineer

Gujarat, India | Jul 2024 – May 2025

- Applied semiconductor process engineering principles in solar cell fabrication, and managed cleanroom processes
- Used Design of Experiments (DOE) and Statistical Process Control (SPC) to troubleshoot production issues and improve yield
- Reduced production cost by optimizing chemical consumption while minimizing cell rejection through better process control

Galamad Aerospace Pte Ltd. | Communication Engineering Intern

Singapore (Remote) | Dec 2023 – Jun 2024

- Designed and fabricated UHF CubeSat antenna prototypes, using HFSS to validate frequency performance at 433 MHz
- Prototyped electro-mechanical deployment circuits and verified burnt-resistor trigger timing and system-level stability